

Lecture 15 Live Problems*Chemical kinetics and equilibrium*

Points toward lab manual pages: **Chem1, Chem2** (section L3 S4)**1. The inversion of simple syrup.**

Let Z be the concentration of sucrose (moles per liter), G the concentration of glucose, and F the concentration of fructose. Suppose $Z(0) = 1.7$ (moles per liter, or molar") and

$$Z' = -0.0002 Z \quad (\text{time in seconds}).$$

- (a) How many seconds before 75% of the sucrose has converted to glucose + fructose?
- (b) What do the change equations and time series look like for F and G ?



2. A reversible reaction.



(a) What does equilibrium look like?

(b) Suppose the parameters are $j = 1$ and $k = 1$. Start with $L = 6$ and $G = 0$. What happens?